Precision Placement in Scaled Dot-Product Attention

Ján Kovačovský

Term Paper Draft ▪ Mathematical Foundations of Machine Learning ▪ 2026

Faculty of Mathematics and Physics, Charles University

Instructor: H. V. Le

Abstract. We study scaled dot-product attention as a precision-placement problem. Rather than asking whether one global low-precision dtype is acceptable, we ask which individual stages of the forward pass can safely run in low precision: storage of the query, key, and value tensors, the logit product QK^T, logit storage, the rowwise softmax, and the final value product. The present draft still analyzes an analytically transparent sketch-based baseline, but its main purpose is to motivate a broader decomposition of attention error into logit perturbation, softmax perturbation, and final output perturbation. The accompanying code now includes exact-attention precision sweeps with policy-level controls and diagnostics for relative logit error, rowwise softmax error, and output error.

Keywords: mixed-precision arithmetic, attention, transformers, numerical stability, floating-point arithmetic

1 Introduction

Scaled dot-product attention [1] is the core computational primitive of transformer architectures. As models grow, the memory and compute cost of the attention forward pass motivates the use of reduced-precision arithmetic. The natural question is a *precision-placement* question: which stages of the forward pass can safely run in low precision, and which still require higher-precision accumulation or reduction?

The relevant stages are: storage of the query, key, and value tensors; the logit product QK^T; logit storage; the rowwise softmax; and the final value product. A global dtype comparison—the usual approach—conflates these stages and cannot identify which *locations* are numerically sensitive. This report instead decomposes the attention pipeline and asks, for each stage separately, whether low precision there produces acceptable error.

The analytical core is Proposition 2, a local perturbation bound for exact attention that separates the effect of logit perturbations (weighted by $\|V\|_2$) from value perturbations (weighted by $\sqrt{n}$ from the row-stochastic spectral norm). This decomposition directly predicts which precision locations are sensitive: logit errors propagate through softmax with unit Lipschitz constant, while value errors propagate with a $\sqrt{n}$ factor that is typically moderate. The sharpest failure mode is therefore low-precision softmax, which can amplify tiny logit perturbations into large probability shifts even when the logit error itself is small.

Sketching-based attention approximations [2] [3] [4] [5] [6] are retained as a methodological baseline: they provide a controlled stress test in which the error decomposition can be verified under a known approximation floor. But they are not the main contribution.

2 Operator-Theoretic Setup

Let $\mathcal{H}_d := \mathbb{R}^d$ and $\mathcal{H}_n := \mathbb{R}^n$, both finite-dimensional Hilbert spaces endowed with their standard Euclidean inner products. We view matrices in $\mathbb{R}^{n\times d}$ as Hilbert–Schmidt operators

$$Q, K, V : \mathcal{H}_d \to \mathcal{H}_n.$$

Their Hilbert–Schmidt norm is denoted by $\|\cdot\|_{\mathrm{HS}}$, while $\|\cdot\|_2$ denotes the spectral norm on linear operators.

The exact scaled logit matrix, viewed as an operator on $\mathcal{H}_n$, is

$$\mathcal{L}(Q,K) := d^{-1/2} QK^* \in \mathcal{L}(\mathcal{H}_n),$$

and the rowwise softmax map is denoted by

$$\Sigma : \mathbb{R}^{n\times n} \to \mathbb{R}^{n\times n}.$$

Accordingly, the attention output is the nonlinear operator

$$\mathcal{A}(Q,K,V) := \Sigma(\mathcal{L}(Q,K))V.$$

3 Analytical Core

The analysis begins with a stability property of the rowwise softmax map, which is used in both the exact-attention and sketching bounds.

Proposition 1 (Rowwise softmax is Hilbert–Schmidt Lipschitz)**.** Let $L, M \in \mathbb{R}^{n\times n}$. Then

$$\|\Sigma(L) - \Sigma(M)\|_{\mathrm{HS}} \le \|L - M\|_{\mathrm{HS}}.$$

Consequently, for every $V \in \mathbb{R}^{n\times d}$,

$$\|\Sigma(L)V - \Sigma(M)V\|_{\mathrm{HS}} \le \|L - M\|_{\mathrm{HS}}\,\|V\|_2.$$

Proof. Let $\sigma : \mathbb{R}^n \to \mathbb{R}^n$ be the softmax map on vectors with Jacobian $D\sigma(x) = \mathrm{Diag}(p) - pp^T$ where $p = \sigma(x)$. For any $z \in \mathbb{R}^n$, $z^T D\sigma(x)\, z = \sum_i p_i z_i^2 - (\sum_i p_i z_i)^2 \le \|z\|_2^2$, so $\|D\sigma(x)\|_2 \le 1$ for all x. The mean value theorem gives $\|\sigma(x) - \sigma(y)\|_2 \le \|x - y\|_2$; applying this row by row and squaring yields the HS bound. The second assertion follows from $\|XV\|_{\mathrm{HS}} \le \|X\|_{\mathrm{HS}}\|V\|_2$. □

3.1 Exact-Attention Perturbation Bound

The next proposition is the main analytical result. It decomposes the attention error into a logit-driven term and a value-driven term, directly predicting the relative sensitivity of different precision placements.

Proposition 2 (Exact-attention local perturbation bound)**.** Let

$$A = \Sigma(L)V, \qquad \widehat{A} = \Sigma(\widehat{L})\widehat{V},$$

where $L, \widehat{L} \in \mathbb{R}^{n\times n}$ and $V, \widehat{V} \in \mathbb{R}^{n\times d}$. Writing

$$\Delta L := \widehat{L} - L, \qquad \Delta V := \widehat{V} - V,$$

one has

$$\|A - \widehat{A}\|_{\mathrm{HS}} \le \|\Delta L\|_{\mathrm{HS}}\,\|V\|_2 + \|\Sigma(\widehat{L})\|_2\,\|\Delta V\|_{\mathrm{HS}}.$$

In particular, since $\Sigma(\widehat{L})$ is row-stochastic and therefore $\|\Sigma(\widehat{L})\|_2 \le \sqrt{n}$,

$$\|A - \widehat{A}\|_{\mathrm{HS}} \le \|\Delta L\|_{\mathrm{HS}}\,\|V\|_2 + \sqrt{n}\,\|\Delta V\|_{\mathrm{HS}}.$$

Proof. Add and subtract $\Sigma(\widehat{L})V$ and apply the triangle inequality:

$$\|A - \widehat{A}\|_{\mathrm{HS}} \le \|\Sigma(L)V - \Sigma(\widehat{L})V\|_{\mathrm{HS}} + \|\Sigma(\widehat{L})(V - \widehat{V})\|_{\mathrm{HS}}.$$

The first term is bounded by Proposition 1:

$$\|\Sigma(L)V - \Sigma(\widehat{L})V\|_{\mathrm{HS}} \le \|L - \widehat{L}\|_{\mathrm{HS}}\,\|V\|_2.$$

For the second,

$$\|\Sigma(\widehat{L})(V - \widehat{V})\|_{\mathrm{HS}} \le \|\Sigma(\widehat{L})\|_2\,\|\Delta V\|_{\mathrm{HS}}.$$

Combining these gives the stated bound. □

Remark 3 (Interpretation for precision placement)**.** Proposition 2 directly predicts the relative danger of different precision choices:

(1) **Logit perturbations** ($\Delta L \neq 0$, $\Delta V = 0$): The error scales as $\|\Delta L\|_{\mathrm{HS}}\|V\|_2$. Since the softmax is 1-Lipschitz (Proposition 1), logit perturbations are *not amplified* by softmax itself. However, when logit perturbations are amplified into probability errors, the amplification ratio ρ_{softmax} can still be large on inputs where the softmax is nearly one-hot—a regime the experiments expose.

(2) **Value perturbations** ($\Delta L = 0$, $\Delta V \neq 0$): The error scales as $\sqrt{n}\,\|\Delta V\|_{\mathrm{HS}}$. This $\sqrt{n}$ factor is moderate for typical sequence lengths but can become significant for very long sequences.

(3) **Low-precision softmax**: When Σ itself is computed in low precision, the Lipschitz bound from Proposition 1 no longer applies, and the effective perturbation of the probability matrix can be much larger than $\|\Delta L\|_{\mathrm{HS}}$. The experiments confirm this: ρ_{softmax} spikes to $\sim 10^3$ on transformer activations when softmax is run in bf16/fp16, while remaining < 1 when only storage is low precision.

3.2 Supporting Background: Sketching as a Controlled Baseline

The following propositions analyze a sketched surrogate $\widetilde{\mathcal{A}}_S(Q,K,V) = \Sigma(d^{-1/2}QSS^*K^*)V$ obtained by replacing the full logit product with a low-dimensional projection through $S \in \mathbb{R}^{d\times s}$ with $s < d$. These results are retained as a methodological baseline: the sketching error provides a known approximation floor against which the finite-precision error can be isolated and studied.

Proposition 4 (Deterministic sketching bound). Let $U \subseteq \mathcal{H}_d$ be the joint row space of Q and K, equivalently the subspace spanned by the columns of $[Q^T \;\; K^T]$, and let $r := \dim U$. Assume that S is an ϵ-subspace embedding for U, meaning

$$\left|\langle S^*u, S^*v\rangle - \langle u, v\rangle\right| \le \epsilon \|u\|_2 \|v\|_2 \qquad \text{for all } u, v \in U.$$

Equivalently, if $U_0 \in \mathbb{R}^{d\times r}$ is any orthonormal basis matrix for U, then $\|U_0^T SS^*U_0 - I_r\|_2 \le \epsilon$. Then

$$\|\mathcal{L}(Q,K) - \widetilde{\mathcal{L}}_S(Q,K)\|_{\mathrm{HS}} \le \frac{\epsilon}{\sqrt{d}} \|Q\|_{\mathrm{HS}} \|K\|_2.$$

Consequently,

$$\|\mathcal{A}(Q,K,V) - \widetilde{\mathcal{A}}_S(Q,K,V)\|_{\mathrm{HS}} \le \frac{\epsilon}{\sqrt{d}} \|Q\|_{\mathrm{HS}} \|K\|_2 \|V\|_2.$$

Proof. Let $U_0 \in \mathbb{R}^{d\times r}$ have orthonormal columns spanning the joint row space of Q and K. Write $Q = Q_U U_0^T$ and $K = K_U U_0^T$, where $Q_U = QU_0 \in \mathbb{R}^{n\times r}$ and $K_U = KU_0 \in \mathbb{R}^{n\times r}$. Then

$$QK^* - QSS^*K^* = Q_U(I_r - U_0^T SS^*U_0)K_U^T.$$

The subspace-embedding condition is exactly $\|U_0^T SS^*U_0 - I_r\|_2 \le \epsilon$, so

$$\|QK^* - QSS^*K^*\|_{\mathrm{HS}} \le \|Q_U\|_{\mathrm{HS}} \|I_r - U_0^T SS^*U_0\|_2 \|K_U^T\|_2 \le \epsilon \|Q\|_{\mathrm{HS}} \|K\|_2,$$

where we used $\|Q_U\|_{\mathrm{HS}} = \|Q\|_{\mathrm{HS}}$ and $\|K_U\|_2 = \|K\|_2$. Multiplying by $d^{-1/2}$ gives the logit bound. The attention-output bound follows from Proposition 1. □

Proposition 5 (Probabilistic subspace embedding guarantee). Let $U \subseteq \mathcal{H}_d$ be an r-dimensional subspace. Let $S \in \mathbb{R}^{d\times s}$ have i.i.d. entries $S_{ij} \sim \mathcal{N}(0, 1/s)$. For $\epsilon \in (0,1)$ and $\delta \in (0,1)$, if

$$s \;\ge\; C\,\frac{r + \ln(1/\delta)}{\epsilon^2},$$

for a universal constant C, then with probability at least $1-\delta$, S is an ϵ-subspace embedding for U.

Proof. Let $U_0 \in \mathbb{R}^{d\times r}$ be a matrix whose columns form an orthonormal basis for U. Then S is an ϵ-subspace embedding for U if and only if $\|U_0^T SS^*U_0 - I_r\|_2 \le \epsilon$. The matrix $U_0^T SS^*U_0 = (S^*U_0)^*(S^*U_0)$ has the same distribution as G^*G/s where $G \in \mathbb{R}^{s\times r}$ has i.i.d. standard normal entries. Standard JL concentration [2] gives

$\Pr(\|G^*G/s - I_r\|_2 > \epsilon) \le 2\exp(-c\,s\epsilon^2 + r)$ for a universal constant $c > 0$, provided s is large enough relative to r/ϵ^2. The stated condition with an appropriate C ensures the failure probability is at most δ. □

Proposition 6 (Finite-precision error for sketched logits)**.** Let $\widehat{\mathcal{L}}_S(Q,K)$ denote the logit matrix computed in floating-point arithmetic with unit roundoff u, and let $\gamma_s := su/(1-su)$ for $su < 1$. Then

$$\|\widetilde{\mathcal{L}}_S(Q,K) - \widehat{\mathcal{L}}_S(Q,K)\|_{\mathrm{HS}} \le \frac{\gamma_s\,\|QS\|_{\mathrm{HS}}\,\|KS\|_{\mathrm{HS}}}{\sqrt{d}}\,.$$

Proof. Write the sketched logit product as $d^{-1/2}AB^*$ with $A = QS \in \mathbb{R}^{n\times s}$ and $B = KS \in \mathbb{R}^{n\times s}$. The standard forward-error bound for matrix multiplication [7] gives

$$|\widehat{AB}^* - AB^*| \le \gamma_s\,|A|\,|B^*|,$$

entrywise in absolute value. Taking the Frobenius (Hilbert–Schmidt) norm and applying the Cauchy–Schwarz inequality for matrices,

$$\|\widehat{AB}^* - AB^*\|_{\mathrm{HS}} \le \gamma_s\,\||A|\,|B^*|\|_{\mathrm{HS}} \le \gamma_s\,\|A\|_{\mathrm{HS}}\,\|B\|_{\mathrm{HS}}.$$

Multiplying by $d^{-1/2}$ yields the bound. □

Remark 7 (Scope of the finite-precision bound)**.** The bound in Proposition 6 models the error in the final sketched logit product $(QS)(KS)^*$, assuming that QS and KS are given exactly. The experiments measure a broader implementation error that also includes storage quantization and low-precision projection effects. ⌟

Combining the subspace embedding guarantee with the finite-precision error yields the sketching main theorem.

Theorem 8 (Mixed-precision sketched attention)**.** Under the hypotheses of Propositions 5 and 6, with probability at least $1-\delta$,

$$\|\mathcal{A}(Q,K,V) - \widehat{\mathcal{A}}_S(Q,K,V)\|_{\mathrm{HS}} \le \frac{1}{\sqrt{d}}\Big(\epsilon\,\|Q\|_{\mathrm{HS}}\,\|K\|_2 + \gamma_s\,\|QS\|_{\mathrm{HS}}\,\|KS\|_{\mathrm{HS}}\Big)\|V\|_2.$$

The first term is the sketching error; the second is the finite-precision error.

Proof. By Proposition 1 and the triangle inequality,

$$\|\mathcal{A} - \widehat{\mathcal{A}}_S\|_{\mathrm{HS}} \le (\|\mathcal{L} - \widetilde{\mathcal{L}}_S\|_{\mathrm{HS}} + \|\widetilde{\mathcal{L}}_S - \widehat{\mathcal{L}}_S\|_{\mathrm{HS}})\,\|V\|_2.$$

Apply Proposition 4 to the first term and Proposition 6 to the second. □

Remark 9 (Design rule from the sketching theorem)**.** Theorem 8 yields a concrete joint design rule. For float16 ($u_{16} \approx 5\times10^{-4}$) and typical sketch dimensions $s \le 64$, we have $s\cdot u_{16} \le 0.032$, which is smaller than typical sketch tolerances $\epsilon \ge 0.1$. For float32 ($u_{32} \approx 6\times10^{-8}$), the condition is essentially always satisfied. For float8 E4M3 ($u_{\mathrm{fp8}} \approx 0.0625$), even $s = 16$ gives $s\cdot u_{\mathrm{fp8}} \approx 1.0$, which is comparable to ϵ—so the finite-precision term is no longer clearly lower order.

4 Numerical Experiments

4.1 Exact Attention Under Precision Policies

For each precision policy, we evaluate the exact attention pipeline

$$Q, K, V \to L = QK^*/\sqrt{d} \to P = \Sigma(L) \to A = PV$$

against a `float64` reference. The primary diagnostics are the relative logit error

$$E_L = \frac{\|L - \widehat{L}\|_{\mathrm{HS}}}{\|L\|_{\mathrm{HS}}},$$

the mean rowwise softmax error

$$E_P = \frac{1}{n} \sum_{i=1}^{n} \|p_i - \widehat{p}_i\|_1,$$

the relative output error

$$E_A = \frac{\|A - \widehat{A}\|_{\mathrm{HS}}}{\|A\|_{\mathrm{HS}}},$$

and the softmax amplification ratio

$$\rho_{\mathrm{softmax}} = \frac{\frac{1}{n}\sum_{i=1}^{n} \|p_i - \widehat{p}_i\|_1}{\frac{1}{n}\sum_{i=1}^{n} \|\ell_i - \widehat{\ell}_i\|_\infty}.$$

We test the following policy family: `fp32_reference`, `bf16_safe`, `fp16_safe`, low-precision logit storage, low-precision softmax, and low-precision value accumulation. The "safe" policies keep storage low precision but preserve fp32 accumulation and fp32 softmax. This separation is the core experimental point: the question is not whether one global dtype works, but which *locations* are sensitive.

Policy	E_L	E_P	E_A	ρ_{softmax}
`fp32_reference`	8.2e-8	8.5e-8	1.4e-7	0.41
`bf16_safe`	2.4e-3	1.9e-3	2.8e-3	0.35
`fp16_safe`	2.9e-4	2.1e-4	3.2e-4	0.32
`bf16_low_logit`	2.9e-3	2.3e-3	3.4e-3	0.34
`fp16_low_logit`	3.5e-4	2.8e-4	4.4e-4	0.33
`bf16_low_softmax`	2.9e-3	2.7e-3	3.7e-3	0.41
`fp16_low_softmax`	3.5e-4	3.3e-4	4.7e-4	0.39
`bf16_low_value`	2.4e-3	1.9e-3	3.5e-3	0.35
`fp16_low_value`	2.9e-4	2.1e-4	4.4e-4	0.32

Table 10 / Exact-attention errors on synthetic Gaussian data ($n \in \{16, 32\}$, $d = 16$, averaged over 2 seeds). "Safe" policies use low-precision storage with fp32 accumulation and softmax.

Tables 10 and 11 report representative policy-level numbers from Gaussian and transformer-activation sweeps respectively. On Gaussian data (Table 10), all policies remain low-error

Policy	E_L	E_P	E_A	ρ_{softmax}
`fp32_reference`	1.8e-8	3.0e-8	5.0e-8	4.6e4
`bf16_safe`	6.1e-4	3.0e-8	1.8e-3	1.9
`fp16_safe`	3.3e-4	3.0e-8	3.5e-4	3.4
`bf16_low_logit`	3.9e-4	3.0e-8	1.8e-3	2.8
`fp16_low_logit`	3.9e-4	3.0e-8	3.5e-4	2.8
`bf16_low_softmax`	3.9e-4	1.2e-5	1.8e-3	1.1e3
`fp16_low_softmax`	3.9e-4	1.2e-5	3.4e-4	1.1e3
`bf16_low_value`	6.1e-4	3.0e-8	1.8e-3	1.9
`fp16_low_value`	3.3e-4	3.0e-8	3.4e-4	3.4

Table 11 / Exact-attention errors on a single attention head extracted from a pretrained transformer (`sshleifer/tiny-distilbert-base-cased`, layer 0, head 0; $n = 16$, $d = 1$). The softmax amplification ratio ρ_{softmax} spikes under low-precision softmax, confirming the prediction from Proposition 2 and Remark 3.

because the softmax is not near-saturated: $\rho_{\text{softmax}} < 1$ uniformly, meaning the softmax actually *damps* logit perturbations. The ordering $E_L < E_P < E_A$ is consistent across all policies, and the differences between "safe" and "low" variants are modest.

The transformer activations (Table 11) reveal a qualitatively different regime. The baseline ρ_{softmax} for `fp32_reference` is $\sim 10^4$ because the denominator (rowwise logit perturbation) is at machine epsilon while the numerator is even smaller; this large ratio is a numerical artifact of dividing by a near-zero baseline. The important comparison is between "safe" policies ($\rho \sim$ 1–3) and "low_softmax" policies ($\rho \sim 10^3$). Low-precision softmax amplifies the tiny logit perturbation ($E_L \approx 4 \times 10^{-4}$) into a softmax error E_P that is two orders of magnitude larger relative to the safe baseline. This is exactly the failure mode predicted by Remark 3(3): when softmax itself is computed in low precision, the Lipschitz guarantee from Proposition 1 no longer holds.

Figure 14 summarizes these policy-level results together with the residual-depth experiment discussed below. The most important empirical message is that low-precision storage of Q, K, V is usually much safer than low-precision softmax or low-precision logit storage. This is precisely the kind of location-specific statement that a global dtype comparison would miss.

4.2 Residual Depth Propagation

To test how local precision errors accumulate across depth, we evaluate an attention-only residual stack of the form

$$x_{\ell+1} = x_\ell + hF_\ell(x_\ell)$$

with self-attention blocks F_ℓ. The metric of interest is the relative terminal-state error

$$E_{\text{depth}} = \frac{\|x_L - \widehat{x}_L\|_{\text{HS}}}{\|x_L\|_{\text{HS}}}.$$

Table 12 reports E_{depth} for both Gaussian and transformer-activation residual stacks. On Gaussian data, error grows modestly with depth for all policies: the fp16-safe error

	Gaussian		Transformer	
Policy	depth 4	depth 8	depth 2	depth 4
`fp32_reference`	7.2e-8	8.8e-8	5.2e-8	2.2e-8
`bf16_safe`	3.3e-3	4.7e-3	3.4e-3	1.1e-3
`fp16_safe`	4.5e-4	6.7e-4	1.6e-4	1.7e-4
`bf16_low_logit`	3.3e-3	4.9e-3	3.4e-3	1.1e-3
`fp16_low_logit`	4.5e-4	6.3e-4	1.6e-4	1.7e-4
`bf16_low_softmax`	3.6e-3	4.7e-3	3.4e-3	1.1e-3
`fp16_low_softmax`	4.3e-4	6.8e-4	1.6e-4	1.7e-4
`bf16_low_value`	3.6e-3	5.3e-3	3.1e-3	1.2e-3
`fp16_low_value`	5.3e-4	7.8e-4	6.6e-4	6.1e-4

Table 12 / Relative terminal-state error E_{depth} for attention-only residual stacks (Gaussian data, $n = d = 16$, averaged over 2 seeds). Depth $\ell = 4$ uses $h = 0.25$; depth $\ell = 8$ uses $h = 0.125$.

roughly doubles from depth 4 to depth 8, while the bf16-safe error grows by roughly 50%. The low-precision value policies show the steepest depth growth, consistent with the $\sqrt{n}\|\Delta V\|_{\text{HS}}$ term in Proposition 2 accumulating across layers. On transformer activations, the residual scaling $h = 1/\text{depth}$ keeps errors approximately constant or even decreasing with depth, reflecting the stabilizing effect of small residual contributions.

4.3 A Stress-Test Baseline

The experiments in this subsection probe the decomposition of Theorem 8 directly. For each combination of parameters (n, d, s) and each precision configuration we compute three logit matrices:

(1) $\mathcal{L}(Q, K)$: exact unsketched logits in `float64` (reference);

(2) $\widetilde{\mathcal{L}}_S(Q, K)$: exact sketched logits in `float64`;

(3) $\widehat{\mathcal{L}}_S(Q, K)$: sketched logits in the target precision.

The sketching error is $E_{\text{sk}} := \|\mathcal{L} - \widetilde{\mathcal{L}}_S\|_{\text{HS}}$ and the finite-precision error is $E_{\text{fp}} := \|\widetilde{\mathcal{L}}_S - \widehat{\mathcal{L}}_S\|_{\text{HS}}$. The total error is $E_{\text{tot}} := \|\mathcal{L} - \widehat{\mathcal{L}}_S\|_{\text{HS}}$. The same decomposition is recorded for the attention output. We test four precision configurations: `fp32` (all stages in float32), `full_fp16` (storage, projection, final logit product, and softmax in float16), `mixed_fp16` (storage and projection in float16, followed by a float32 final logit product and float32 softmax), and `fp8_store` (storage and projection in float8 E4M3, followed by a float32 final logit product and float32 softmax).

The sketch $S \in \mathbb{R}^{d \times s}$ is a scaled Gaussian matrix with i.i.d. $\mathcal{N}(0, 1/s)$ entries; the scaling is chosen so that $\mathbb{E}[SS^*] = I_d$. We sweep $n \in \{64, 128, 256\}$, $d \in \{32, 64\}$, $s \in \{8, 16, 24, 32\}$, and average over three seeds. Data and sketch seeds are independent. The full numerical results are available in the committed CSV file `results/attention_sweep.csv`.

Since the synthetic Q and K are i.i.d. Gaussian, their joint row space is typically full-dimensional, so for $d = 64$ and $s \leq 32$ the subspace-embedding regime from Proposition 5 is not expected to hold. The experiment should therefore be read as an intentionally under-sketched stress test of the error decomposition, not as evidence that such small feature sketches preserve attention accurately.

Table 13 reports absolute HS errors of the logits for $n = 256$, $d = 64$, together with the ratio $E_{\text{sk}}/E_{\text{fp}}$; the output errors showed the same qualitative ordering and are omitted.

Config	s	E_{sk}	E_{fp}	E_{tot}	$E_{\text{sk}}/E_{\text{fp}}$
`fp32`	8	7.22e2	1.2e-4	7.22e2	5.9e6
	16	5.03e2	1.1e-4	5.03e2	4.4e6
	24	4.27e2	9.4e-5	4.27e2	4.5e6
	32	3.70e2	9.1e-5	3.70e2	4.1e6
`full_fp16`	8	7.22e2	4.1e-1	7.22e2	1.8e3
	16	5.03e2	2.9e-1	5.03e2	1.8e3
	24	4.27e2	2.6e-1	4.27e2	1.7e3
	32	3.70e2	2.3e-1	3.70e2	1.6e3
`mixed_fp16`	8	7.22e2	3.8e-1	7.22e2	1.9e3
	16	5.03e2	2.6e-1	5.03e2	1.9e3
	24	4.27e2	2.4e-1	4.27e2	1.8e3
	32	3.70e2	2.1e-1	3.70e2	1.8e3
`fp8_store`	8	7.22e2	5.0e1	7.21e2	1.4e1
	16	5.03e2	3.4e1	5.01e2	1.5e1
	24	4.27e2	3.0e1	4.26e2	1.4e1
	32	3.70e2	2.6e1	3.70e2	1.4e1

Table 13 / Hilbert–Schmidt errors of the logit matrix ($n = 256$, $d = 64$, averaged over 3 seeds). $E_{\text{sk}} = \|\mathcal{L} - \widetilde{\mathcal{L}}_S\|_{\text{HS}}$; $E_{\text{fp}} = \|\widetilde{\mathcal{L}}_S - \widehat{\mathcal{L}}_S\|_{\text{HS}}$; $E_{\text{tot}} = \|\mathcal{L} - \widehat{\mathcal{L}}_S\|_{\text{HS}}$; $\|\mathcal{L}\|_{\text{HS}} \approx 256$.

Three observations stand out. First, for fp32 and fp16, $E_{\text{sk}}/E_{\text{fp}}$ is consistently large (10^3–10^6), confirming that the dominant error source is the randomized sketch. Second, `full_fp16` and `mixed_fp16` have finite-precision errors of the same order, consistent with Proposition 6: storage quantization dominates any benefit from higher-precision accumulation. Third, fp8 storage substantially increases the finite-precision error (from $\sim$0.3 to $\sim$30 in absolute terms), but the sketching error still dominates by a factor of $\sim$14–15. The results support the error decomposition and illustrate the degradation predicted by Theorem 8 for increasing $s \cdot u$, but they do not demonstrate a high-quality efficient-attention approximation or a full inversion of the error ratio.

5 Outlook

The main contribution of this report is the precision-placement decomposition of Proposition 2 and its empirical validation: low-precision softmax is the clearest failure mode in exact attention, while low-precision storage of Q, K, V with fp32 accumulation is typically safe. The sketching baseline (Theorem 8) confirms that the finite-precision term remains lower order as long as $s \cdot u \ll \epsilon$.

Several directions remain open. First, the current experiments use small problem sizes ($n \leq 256$, $d \leq 64$); scaling to realistic transformer dimensions will require either GPU acceleration or the JAX backend scaffold already present in the repository. Second,

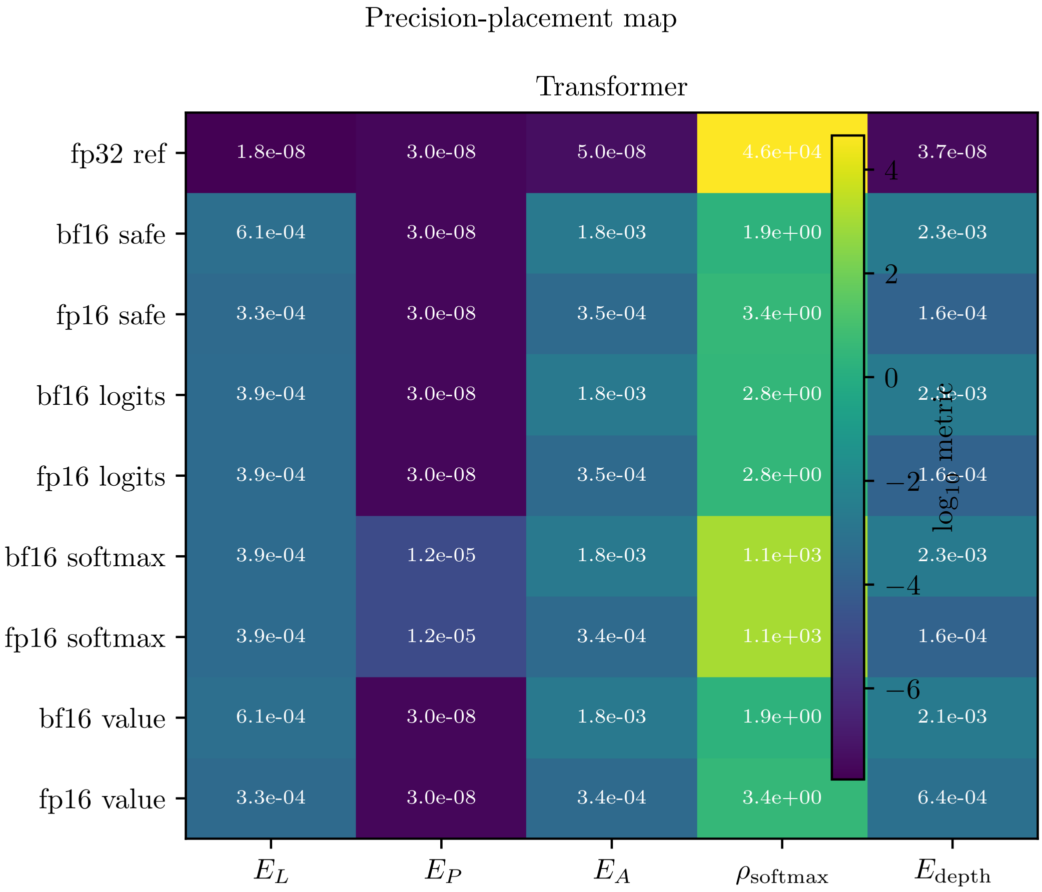

Figure 14 / Precision-placement map for exact attention. Rows correspond to explicit precision policies, columns to the principal diagnostics E_L, E_P, E_A, ρ_{softmax}, and E_{depth}. The Gaussian panel gives the synthetic baseline; the transformer panel uses a fixed head from a pretrained model. Safe policies with fp32 accumulation and fp32 softmax remain low-error, whereas low-precision softmax is the clearest failure mode through its sharp increase in ρ_{softmax}.

FP8 policies are reported only under an explicit quantization model; a more complete treatment would include hardware-specific scaling and dequantization. Third, the phase-diagram analogy from Decelle et al. [8] suggests mapping not only average error but also regime changes: when does low-precision logit storage remain harmless, when does low-precision softmax begin to amplify, and when does no practical low-precision policy preserve the reference behavior? The current data already hints at these transitions but does not yet chart them systematically. Fourth, the JAX-focused mixed-precision framework of Graefe and Trimpe [9] provides a natural implementation path for the safe/unsafe policy distinctions identified here.

On the analytical side, the $\sqrt{n}$ bound on $\|\Sigma(\widehat{L})\|_2$ is likely loose for near-doubly-stochastic attention matrices; a refined bound that exploits the concentration of softmax rows could tighten the value-perturbation term. The logit-perturbation term could similarly be refined by replacing the worst-case $\|V\|_2$ with an average-case or data-dependent quantity.

References

Back-references to the pages where the publication was cited are given by [•].

[1] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention Is All You Need. In: *Advances in Neural Information Processing Systems 30*. **2017**.
arXiv: 1706.03762 [cs.CL] [1]

[2] Nathan Halko, Per-Gunnar Martinsson, and Joel A. Tropp. Finding Structure with Randomness: Probabilistic Algorithms for Constructing Approximate Matrix Decompositions. *SIAM Review*, **2011**.
DOI: 10.1137/090771806 [2], [4]

[3] Michael W. Mahoney. Lecture Notes on Randomized Linear Algebra. **2016**.
arXiv: 1608.04481 [cs.DS] [2]

[4] Sinong Wang, Belinda Z. Li, Madian Khabsa, Han Fang, and Hao Ma. Linformer: Self-Attention with Linear Complexity. **2020**.
arXiv: 2006.04768 [cs.LG] [2]

[5] Yunyang Xiong, Zhanpeng Zeng, Rudrasis Chakraborty, Mingxing Tan, Glenn Fung, Yin Li, and Vikas Singh. Nyströmformer: A Nyström-Based Algorithm for Approximating Self-Attention. **2021**.
arXiv: 2102.03902 [cs.CL] [2]

[6] Krzysztof Choromanski, Valerii Likhosherstov, David Dohan, Xingyou Song, Andreea Gane, Tamas Sarlos, Peter Hawkins, Jared Davis, Afroz Mohiuddin, Lukasz Kaiser, David Belanger, Lucy Colwell, and Adrian Weller. Rethinking Attention with Performers. **2021**.
arXiv: 2009.14794 [cs.LG] [2]

[7] Nicholas J. Higham. Accuracy and Stability of Numerical Algorithms. 2nd ed. SIAM, **2002**.
DOI: 10.1137/1.9780898718027 [5]

[8] Aurelien Decelle, Florent Krzakala, Cristopher Moore, and Lenka Zdeborová. Phase Transition in the Detection of Modules in Sparse Networks. *Physical Review Letters*, **2011**.
DOI: 10.1103/PhysRevLett.107.065701 [10]

[9] Alexander Gr"afe and Sebastian Trimpe. MPX: Mixed Precision Training for JAX. **2025**.
arXiv: 2507.03312 [cs.LG] [10]